

Adam Shaw

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First author on three papers in scientific machine learning and representation learning, spanning gauge-equivariant neural networks, learned preconditioning for iterative solvers, and topological analysis of neural representations. Currently enrolled in a joint B.S./M.S. program with plans to apply to CS Ph.D. programs.

EDUCATION

University of Southern California

Los Angeles, CA

M.S. in Statistics

Expected December 2027

B.S. in Applied and Computational Mathematics, Minor in Computer Science

Expected May 2027

Undergraduate GPA 3.7 • Graduate GPA 4.0

University of Edinburgh (Semester Abroad)

Spring 2026

PUBLICATIONS

- [1] **Adam Shaw**, Yixuan Sun, Srinivas Eswar, William Detmold, Phiala Shanahan, Xiaoye Li, and Yang Liu. “[Learning Lattice Variables for Matrix-Free Wilson-Dirac Preconditioning in Lattice QCD](#).” *Preprint, 2026*. Under review at the [NeurIPS 2026 Workshop on Representations for the Physical Sciences](#).
- [2] **Adam Shaw**, Jiayu Li, Michael Sperling, Michael Kim, and Alvin Jin. “[Topological Simplification in Predictive Coding Networks](#).” *Proceedings of the 2nd Conference on Topology, Algebra, and Geometry in Data Science (TAG-DS 2026)*, PMLR 334(2):316–332, 2026.
- [3] **Adam Shaw**, Jiayu Li, Michael Sperling, Michael Kim, and Alvin Jin. “[Symmetry Acquisition in Predictive Coding Networks](#).” *ICML 2026 Workshop on Weight-Space Symmetries*. Accepted (non-archival).

RESEARCH EXPERIENCE

Argonne National Laboratory (SULI Intern)

Summer 2026

Equivariant deep learning for large-scale numerical solvers

Lemont, IL

- Designed **matrix-free gauge-equivariant neural preconditioners** that learn to accelerate the iterative solution of the Dirac equation in lattice quantum chromodynamics (LQCD).
- Achieved **zero-shot transfer across lattice volumes**, with models trained on smaller lattices matching or exceeding their own speedup when run on larger ones, whereas the SOTA degraded without re-tuning.
- Built the full multi-GPU data-parallel training and evaluation pipeline in JAX, **training end-to-end** through **unrolled iterative solver iterations**.
- Proposed three novel architectures and diagnosed why two failed to improve results, tracing the outcome to specific structural properties and identifying directions for future improvement.

- Studied how **learned representations simplify layer-by-layer** in invertible neural networks and how that simplification affects a network’s ability to invert; "simplification" was measured via topological invariants.
- Built an extensive Python research codebase implementing predictive coding networks in JAX and a representation-analysis stack in Ripser, NumPy, and scikit-learn.
- Produced two accepted papers and two conference posters, one chosen for a lightning talk (TAG-DS).

INDUSTRY EXPERIENCE

Google — Associate Software Developer Intern

Summer 2025

Workspace AI Platform — Gemini integration across Docs, Slides, Gmail, and more

New York, NY

- Designed and built an access control system for custom Gems (personalized Gemini instances), streamlining creation and secure sharing across Google Workspace teams.
- Implemented role-based permissions in C++ on Google’s internal server framework, handling authorization and routing for millions of potential users.
- Automated enum ID generation across a 10,000+ line shared codebase in Python, eliminating merge conflicts for a multi-engineer team.

Rollick — Software Engineer Intern

Summer 2024

Front-end development and campaign personalization

Chicago, IL

- Built modular front-end templates in Nunjucks and jQuery and optimized UTM-based personalization, working in a standard review-and-merge Git workflow.

AWARDS AND HONORS

USC Presidential Scholarship

2023

National Merit Scholarship Winner

2023

TECHNICAL SKILLS

Languages	Python, C++, JavaScript
ML & scientific computing	JAX, NumPy, Pandas, scikit-learn, Ripser, Matplotlib
Methods	Equivariant and invertible neural architectures, automatic differentiation through iterative solvers, numerical linear algebra and Krylov solvers, persistent homology
Tools	Git/GitHub, Piper, Blaze, Protobuf, gRPC, HPC clusters (PBS), Linux, Bash, LaTeX

RELEVANT COURSEWORK

Undergraduate	Machine Learning, Mathematics of ML, NLP, Mathematical Statistics, Real Analysis
Graduate	Analysis of Algorithms, Numerical Methods, Applied Probability